

SoftMarginLoss#

<https://pytorch.org/docs/stable/generated/torch.nn.SoftMarginLoss.html#torch>

Description:

Creates a criterion that optimizes a two-class classification logistic loss between input tensor `xxx` and target tensor `yyy` (containing 1 or -1).

`size_average` (bool, optional) – Deprecated (see `reduction`). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field `size_average` is set to `False`, the losses are instead summed for each minibatch. Ignored when `reduce` is `False`. Default: `True`

`reduce` (bool, optional) – Deprecated (see `reduction`). By default, the losses are averaged or summed over observations for each minibatch depending on `size_average`. When `reduce` is `False`, returns a loss per batch element instead and ignores `size_average`. Default: `True`

`reduction` (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: `size_average` and `reduce` are in the process of being deprecated, and in the meantime, specifying either of those two args will override `reduction`. Default: 'mean'

Input: $(*)$ $(*)$ $(*)$, w

Function Signature

```
class torch.nn.SoftMarginLoss(size_average=None,  
reduce=None, reduction='mean')[source]#
```

Parameters

Shape:

```
forward(input, target)[source]#
```

Return type

Returns:

Tensor

Detailed Documentation

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Input: $(*)^{(*)}(*_*)$, where $*_*$ means any number of dimensions.

Target: $(*)^{(*)}(*_*)$, same shape as the input.

Output: scalar. If reduction is 'none', then $(*)^{(*)}(*_*)$, same shape as input.

Runs the forward pass.

